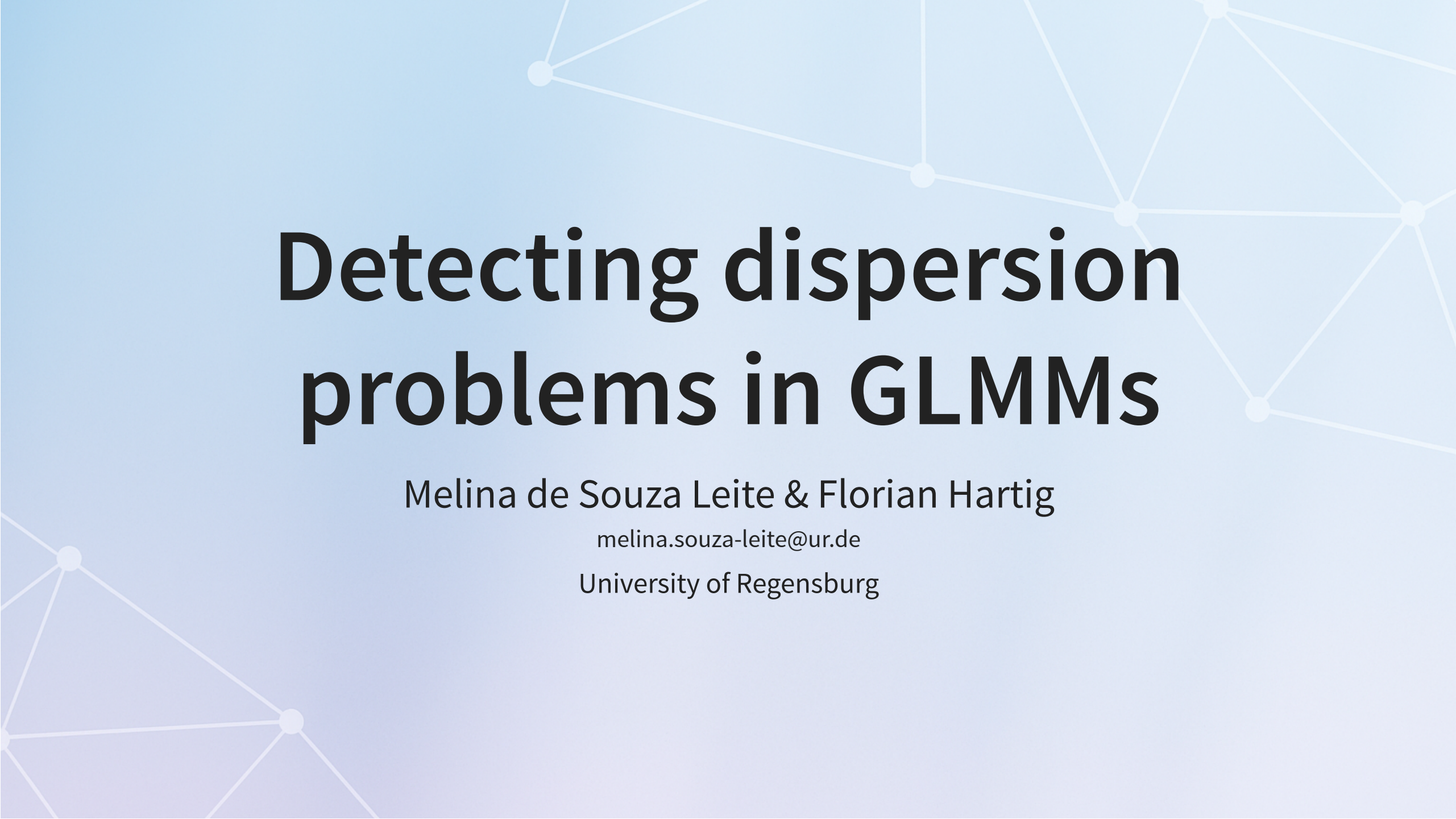




Detecting dispersion problems in GLMMs

Melina de Souza Leite & Florian Hartig

melina.souza-leite@ur.de

University of Regensburg

Replicability crisis in Ecology too?

- High **false positive rates** (type I error) in Ecological studies
- A non-intentional cause is **failing in checking model's assumptions**
- Researchers **don't check models** -> higher chances of false conclusions
- Few tools for **model diagnostics** of GLMs, GLMMs.



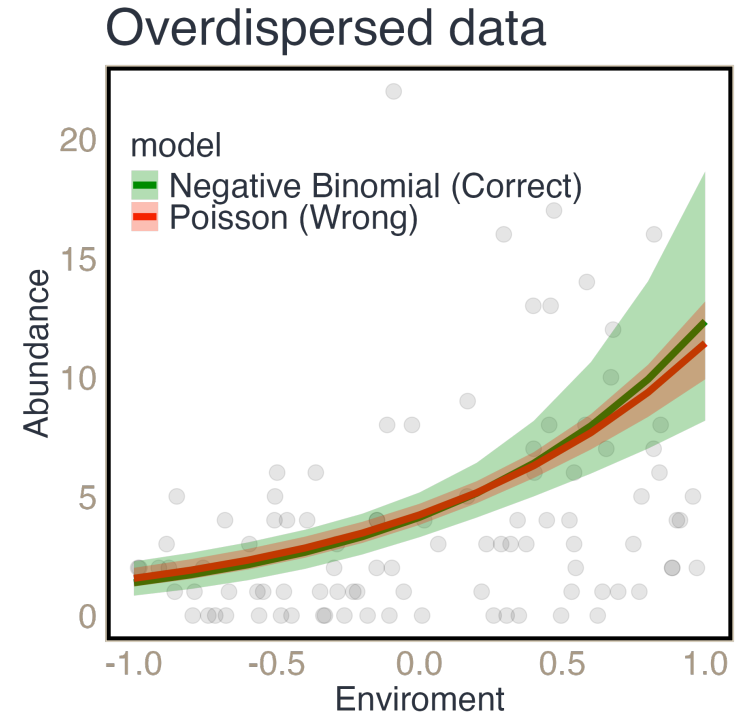
Can you trust your model?

OVERDispersion problems

When data has **more variability than expected** by the model (GLM/GLMM)

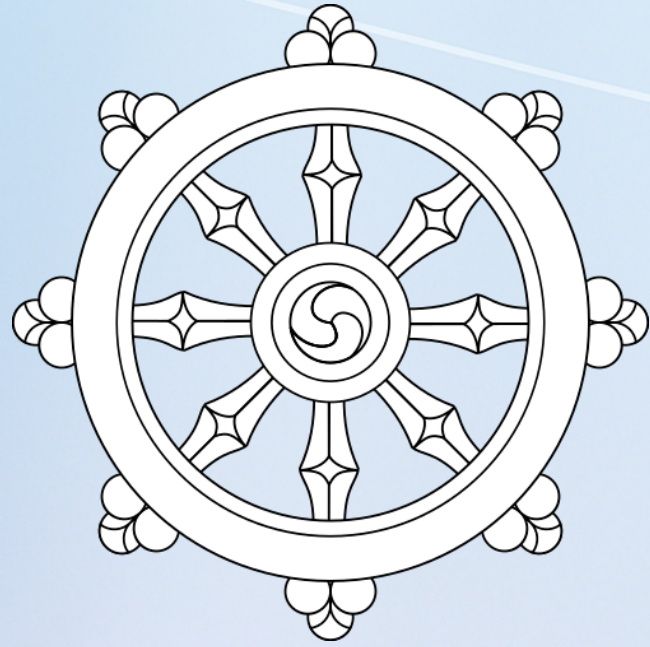
OVERDISPERSION

- Too small standard error / confidence intervals
- Larger chance of **false positive** results
- Much more common than previously thought



GOALS

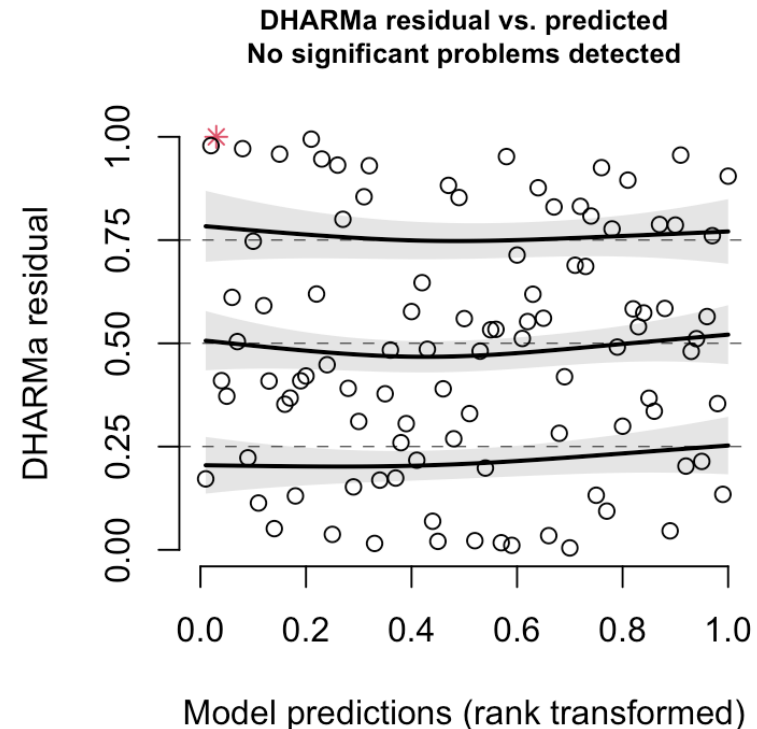
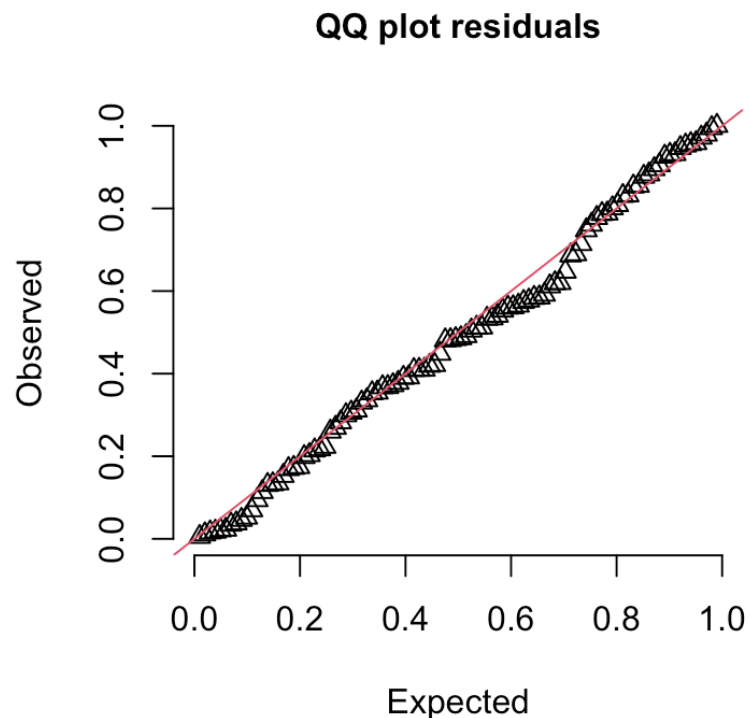
- Aware ecologists of dispersion problems in GLMs/GLMMs
- Identify and describe the 3 main causes by using model diagnostics tools with the [DHARMa](#) R package
- Show some modeling solutions for these causes, with the [glmmTMB](#) R package



Detecting dispersion problems with **DHARMa**

Residual diagnostics with DHARMa

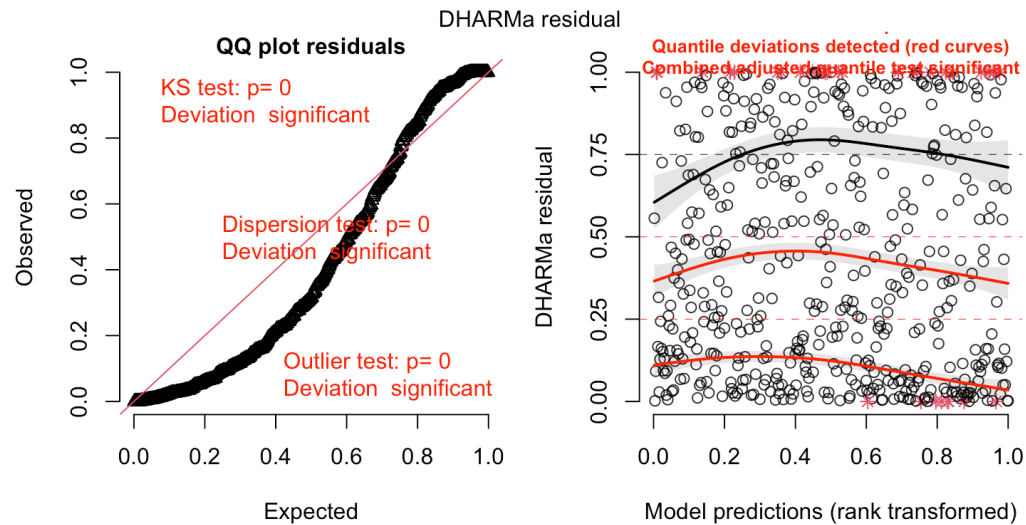
- Scaled quantile residuals -> Simulating from the model
- Residuals between 0 and 1 for ANY model complexity or distribution
- Interpreted the SAME way:



Detecting overdispersion

Wrong model

```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plot(res)  
5 testDispersion(res)
```

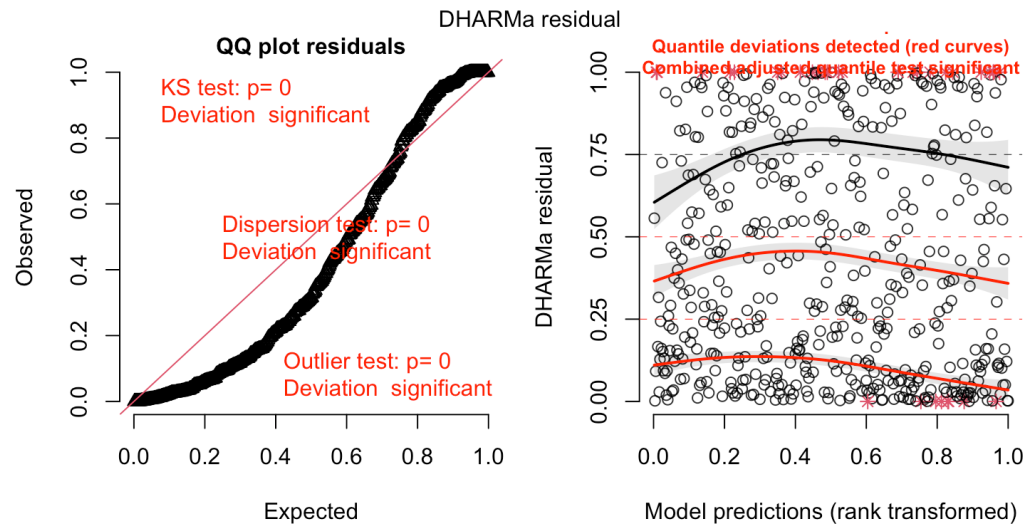


Dispersion = 5.19, p-value = 0.

Detecting overdispersion

Wrong model

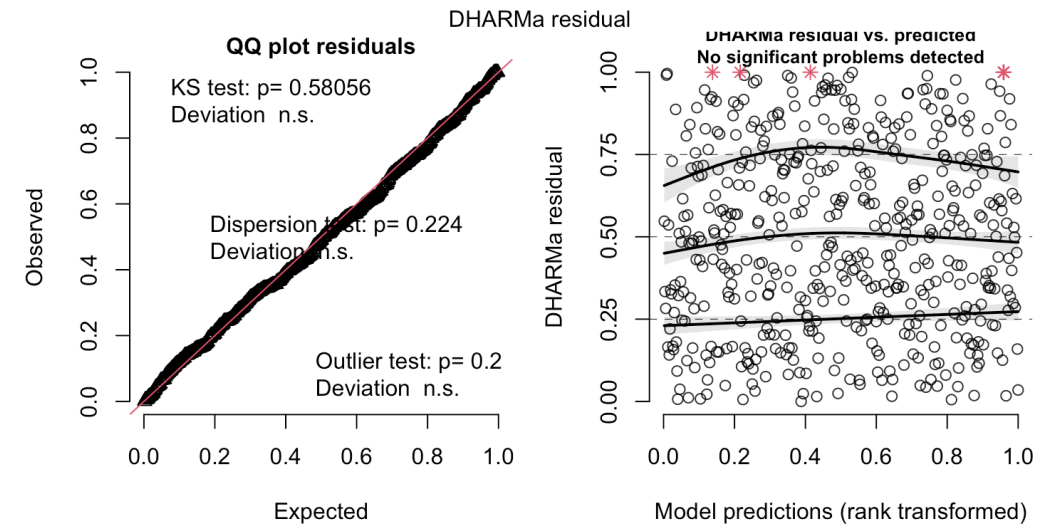
```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plot(res)  
5 testDispersion(res)
```



Dispersion = 5.19, p-value = 0.

Solution

```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = nbinom2(), data = data)  
3 res <- simulateResiduals(m)  
4 plot(res)  
5 testDispersion(res)
```



Dispersion = 1.19, p-value = 0.224.

Another overdispersion problem

What if...

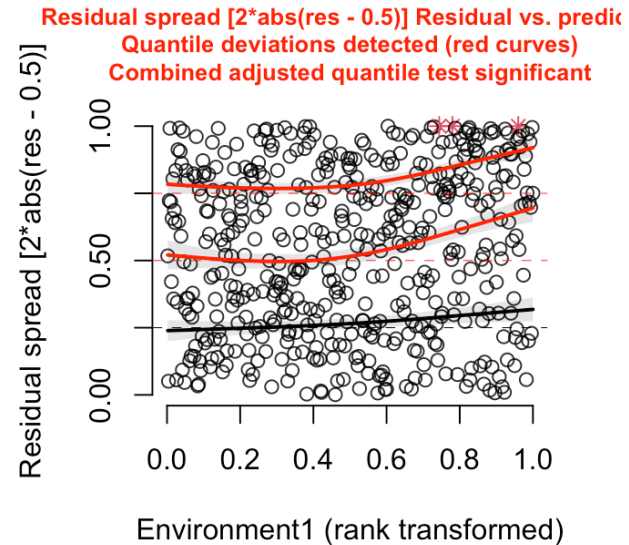
the variability in the data increases/decreases with a specific variable?

Heteroscedasticity

Detecting heteroscedasticity

Wrong model

```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plotResiduals(res, form = data$Environment1,  
5             absoluteDeviation = T)  
6 testDispersion(res)
```

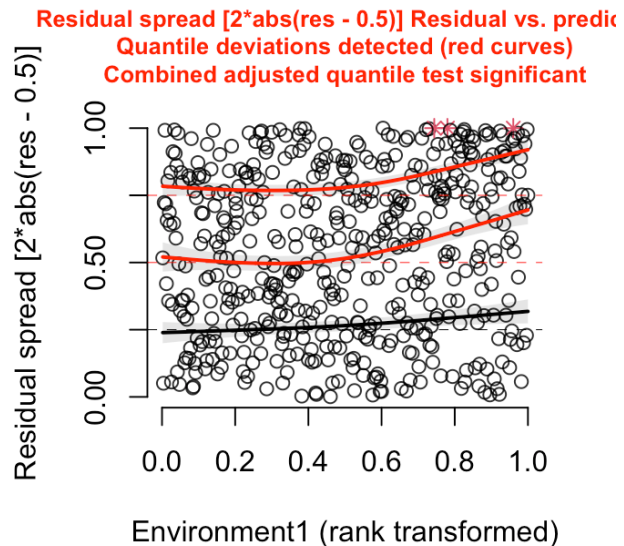


Dispersion = 1.9, p-value = 0.

Detecting heteroscedasticity

Wrong model

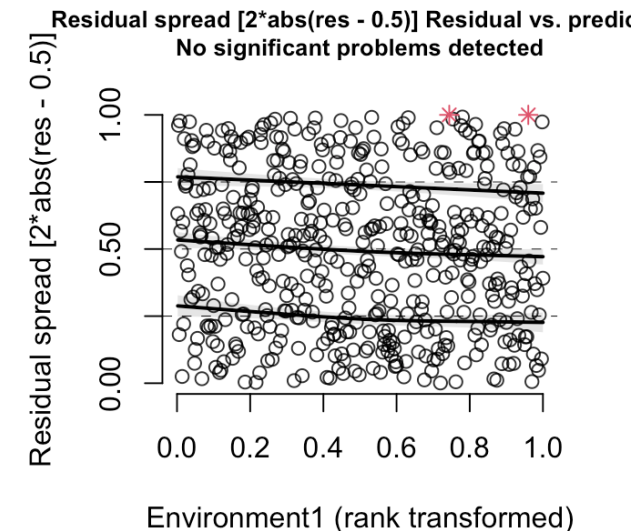
```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plotResiduals(res, form = data$Environment1,  
5               absoluteDeviation = T)  
6 testDispersion(res)
```



Dispersion = 1.9, p-value = 0.

Solution

```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             dispformula = ~ Environment, # dispersion formula  
3             family = nbinom2(), data = data) # negative binomial  
4 res <- simulateResiduals(m)  
5 plotResiduals(res, form = data$Environment1,  
6               absoluteDeviation = T)  
7 testDispersion(res)
```



Dispersion = 1.11, p-value = 0.44.

What if...

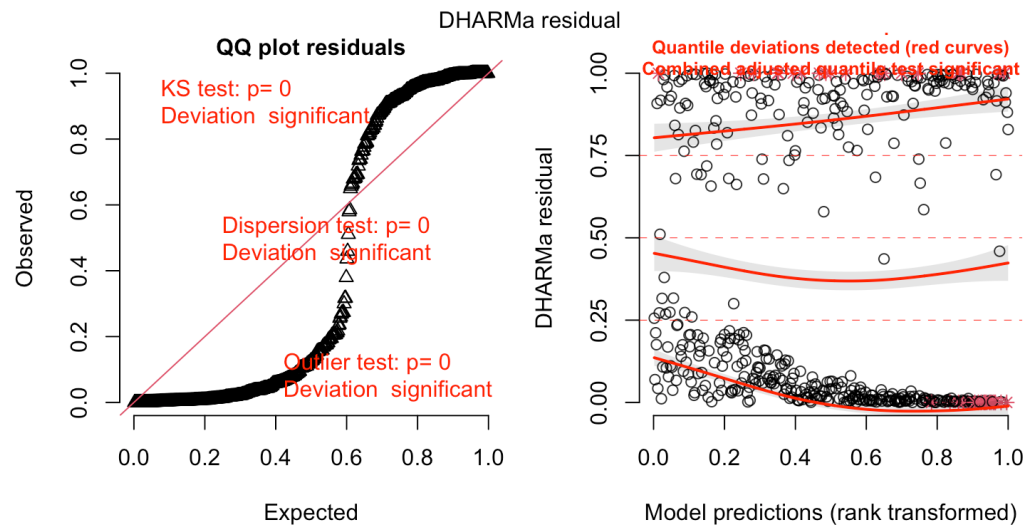
You have more zeros than expected by the model??

Zero-inflation

Detecting zero-inflation

Wrong model

```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plot(res)  
5 testZeroInflation(res)
```

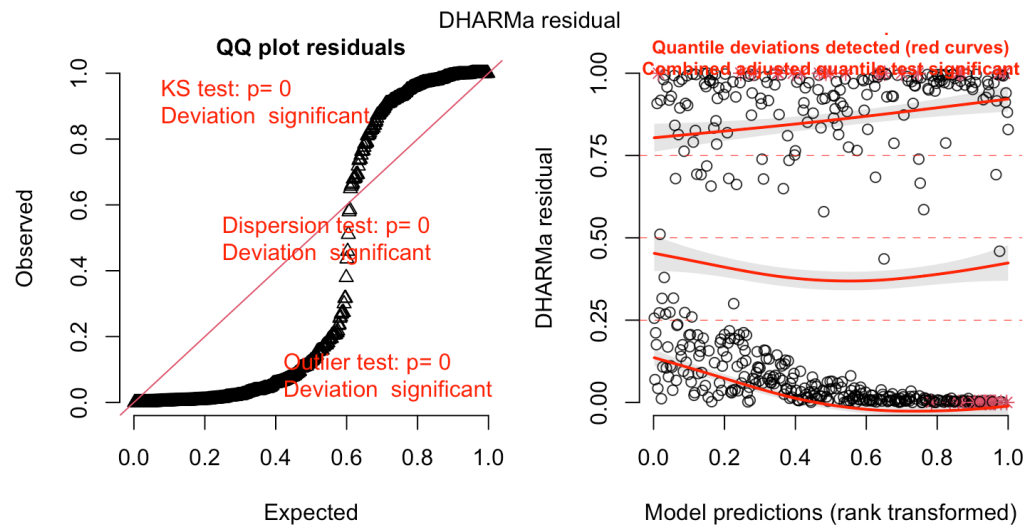


Zero-inflation = 5.16, p-value = 0.

Detecting zero-inflation

Wrong model

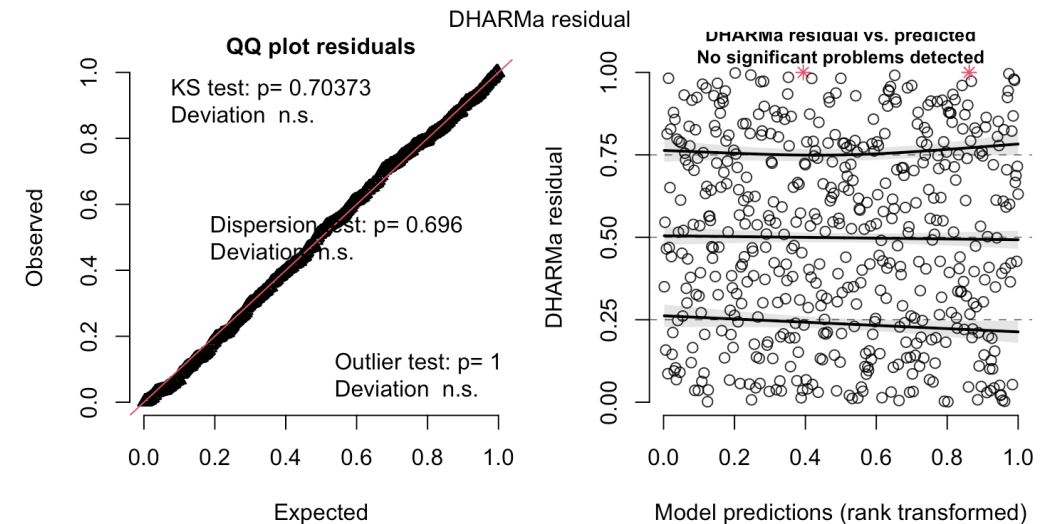
```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             family = poisson(), data = data)  
3 res <- simulateResiduals(m)  
4 plot(res)  
5 testZeroInflation(res)
```



Zero-inflation = 5.16, p -value = 0.

Solution

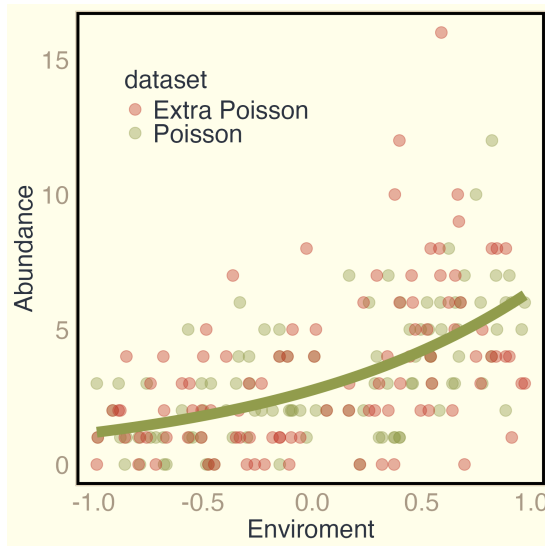
```
1 m <- glmmTMB(Abundance ~ Environment + (1|group),  
2             ziformula = ~ 1, # zero-inflation formula  
3             family = poisson(), data = data)  
4 res <- simulateResiduals(m)  
5 plotResiduals(res)  
6 testZeroInflation(res)
```



Zero-inflation = 1, p -value = 1.

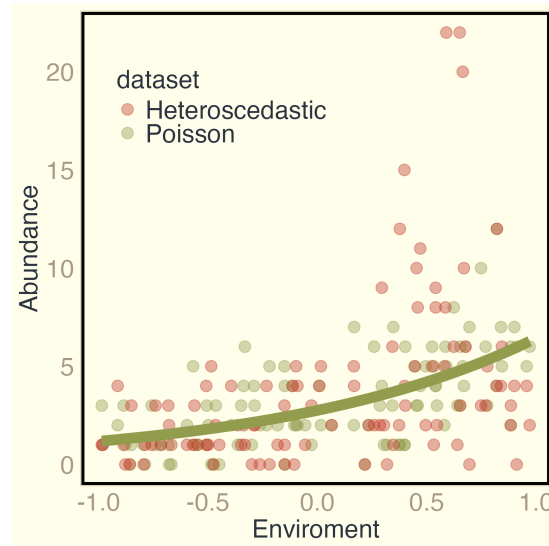
3 causes of dispersion problems

“Real” overdispersion:



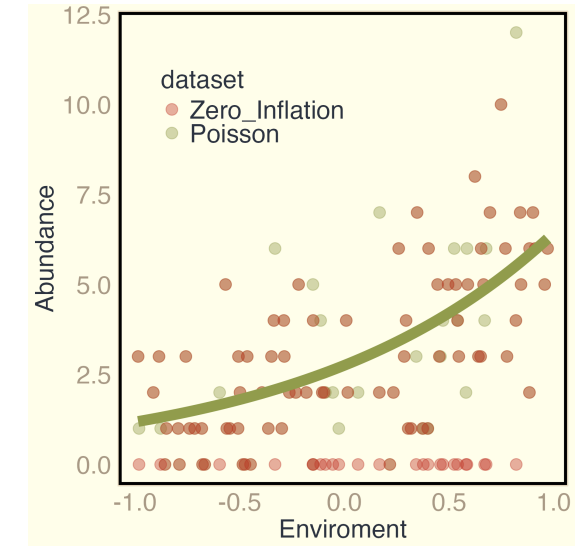
Abundances vary more than expected by the model, in general.

Heteroscedasticity:



Abundance variation increases with the environmental gradient.

Zero-inflation:



More zero abundances than expected by the model.

Detecting dispersion problems

- Residual patterns alone will not tell you which is the cause of overdispersion. E.g.:
 - ‘Real’ overdispersion will show significant test for zero-inflation, and vice-versa.
 - ‘Real’ overdispersion and zero-inflation may have significant heteroscedasticity.
- Additional check: fit models addressing the potential problems and compare their fit (e.g. AIC, LRT) and residuals diagnostics.

DHARMa + glmmTMB

Conclusion

- There are many causes of dispersion problems in GLMMs. To ignore it may lead to false conclusions
- Use [DHARMa](#) residuals tools to detect them
- Address the problem with adequate models, e.g, [glmmTMB](#)

Can you trust your model, now?

Take-home message

- Models should ALWAYS be checked: residual diagnostics!
- Avoid an oversimplistic view of dispersion problems
- Detecting and addressing the causes of dispersion problems may also be informative for your system/data.

Comming soon:

Leite et al. *in prep.* **Dispersion tests in GLMMs: a methods comparison and practical guide.**

Thank you!



Vielen Dank!

Thanks to Max Pichler and the Theoretical Ecology Lab group